

1.Project Title:

Insurance Charges Prediction Using Machine Learning

2.Problem Statement:

Insurance companies need accurate methods for estimating medical expenses and insurance premiums. Manual calculations and fixed rules may not provide precise predictions.

The problem addressed in this project is:

To develop a machine learning system capable of predicting insurance charges based on customer demographic and health-related information while providing business insights through interactive dashboards.

3.Objective:

- To perform Exploratory Data Analysis using Python.
- To identify factors affecting insurance charges.
- To develop a Linear Regression prediction model and evaluate model performance.
- To create an interactive Power BI dashboard.
- To estimate insurance charges for new customers.

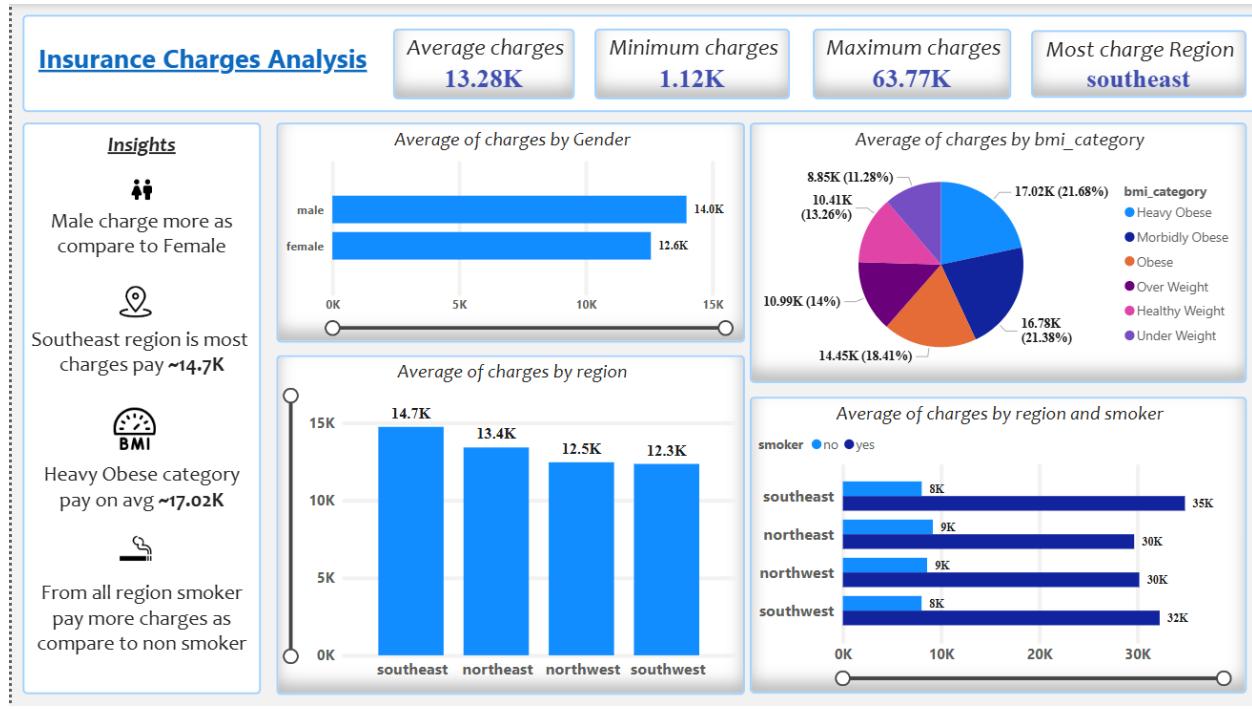
4. Dataset Description:

- Age - Age of the Customer
- Gender - Gender of the Customer (Male/Female)
- BMI - Body Mass Index
- Children - Number of children
- Smoker - Smoking status (Yes/ No)
- Region - Residential region
- Charges - Insurance charges (Target Variable)

5. Workflow:

1. The insurance dataset was collected and imported into Python. It contains customer information such as age, sex, BMI, children, smoker status, region, and insurance charges.
2. EDA was performed using Python libraries like Pandas, Matplotlib, and Seaborn. Various statistical analyses and visualizations were created to identify patterns and factors affecting insurance charges.
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4. Categorical features were encoded, BMI categories were created, and MinMaxScaler was applied to numerical features. This step prepared the data for machine learning model training.
5. The Linear Regression algorithm was used to train the prediction model. The dataset was divided into training and testing sets to build and validate the model.
6. The model was evaluated using R^2 Score, MAE, MSE, and RMSE metrics. After achieving satisfactory performance, the trained model and scaler were saved using Pickle.
7. A Python-based code was developed to accept user inputs such as age, BMI, smoker status, and region. The application processes the inputs and predicts the estimated insurance charges using the trained model.

6. Dashboard



7.Insight:

- Smokers pay nearly 3–4 times higher charges than non-smokers across all regions, indicating that smoking significantly increases health-related risks and medical expenses.
- The Southeast region records the highest average charges (₹14.7K), suggesting a higher concentration of high-risk customers or healthcare expenses in this region.
- Customers in the Heavy Obese and Morbidly Obese categories pay substantially more than those with a healthy BMI, showing a strong relationship between obesity and medical expenditure.
- While average charges vary by region, smoking status and BMI have a much stronger influence on insurance costs than location.
- Individuals who are smokers and belong to obese BMI categories represent the highest-risk group for insurance providers.

